

OPEN SOURCE TECHNOLOGIES (01CE0618) Lab Manual

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Batch: A

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Experiment 2

AIM: Setup Git and Explore Commands Related to Version Control System (VCS)

```
Z:\>set home=c:\users\student
```

```
Z:\>c:
```

```
C:\>git config --global user.name"chandanktech1"
```

```
C:\>git config --global user.email"chandanktech07@gmail.com"
```

```
C:\>git --version
```

```
git version 2.45.1.windows.1
```

```
C:\>git init
```

```
Reinitialized existing Git repository in C:/./git/
```

```
C:\>mkdir ost
```

```
A subdirectory or file ost already exists.
```

```
C:\>mkdir ost
```

```
A subdirectory or file ost already exists.
```

```
C:\>git init
```

```
Reinitialized existing Git repository in C:/./git/
```

```
C:\>cd ost
```

```
C:\ost>git init
```

```
Reinitialized existing Git repository in C:/ost/.git/
```

```
C:\ost>git status
```

```
On branch master
```

```
No commits yet
```

```
nothing to commit (create/copy files and use "git add" to track)
```

```
C:\ost>echo sample.txt
```

```
sample.txt
```

```
C:\ost>echodemo.py
```

```
'echodemo.py' is not recognized as an internal or external command,
```

operable program or batch file.

```
C:\ost>echo demo.py
```

demo.py

```
C:\ost>git status
```

On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

```
C:\ost>dir
```

Volume in drive C has no label.

Volume Serial Number is 95C3-F9A6

Directory of C:\ost

29-01-2026 12:16 <DIR> .

29-01-2026 12:16 <DIR> ..

0 File(s) 0 bytes

2 Dir(s) 74,900,779,008 bytes free

```
C:\ost>echo sample.txt
```

sample.txt

```
C:\ost>dir
```

Volume in drive C has no label.

Volume Serial Number is 95C3-F9A6

Directory of C:\ost

29-01-2026 12:16 <DIR> .

29-01-2026 12:16 <DIR> ..

0 File(s) 0 bytes

2 Dir(s) 74,900,844,544 bytes free

```
C:\ost>echo Hello> sample.txt
```

```
C:\ost>dir
```

Volume in drive C has no label.

Volume Serial Number is 95C3-F9A6

Directory of C:\ost

```
29-01-2026 12:37 <DIR>      .
29-01-2026 12:37 <DIR>      ..
29-01-2026 12:37          7 sample.txt

1 File(s)          7 bytes
2 Dir(s) 74,900,303,872 bytes free
```

C:\ost>type sample.txt

Hello

C:\ost>git commit -m "Initial Commit"

On branch master

Initial commit

Untracked files:

(use "git add <file>..." to include in what will be committed)

sample.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\ost>git add sample.txt

C:\ost>git commit -m "Initial Commit"

[master (root-commit) 4db9386] Initial Commit

Committer: Students <student@mefgi.com>

Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.

You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

git config --global --edit

After doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

1 file changed, 1 insertion(+)

create mode 100644 sample.txt

C:\ost>git branch

* master

C:\ost>git branch -m master main

'git' is not recognized as an internal or external command,

operable program or batch file.

C:\ost>git branch

* master

C:\ost>git checkout main

error: pathspec 'main' did not match any file(s) known to git

C:\ost>git checkout -b dev

Switched to a new branch 'dev'

C:\ost>echo Hello from dev>>sample.txt

C:\ost>git add .

C:\ost>git commit -m "Update in dev"

[dev 33b4787] Update in dev

Committer: Students <student@mefgi.com>

Your name and email address were configured automatically based

on your username and hostname. Please check that they are accurate.

You can suppress this message by setting them explicitly. Run the

following command and follow the instructions in your editor to edit

your configuration file:

git config --global --edit

After doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

1 file changed, 1 insertion(+)

C:\ost>git merge dev

Already up to date.

C:\ost>dir

Volume in drive C has no label.

Volume Serial Number is 95C3-F9A6

Directory of C:\ost

29-01-2026 12:37 <DIR> .

29-01-2026 12:37 <DIR> ..

29-01-2026 12:41 23 sample.txt

1 File(s) 23 bytes

2 Dir(s) 74,907,422,720 bytes free

C:\ost>type sample.txt

Hello

Hello from dev

C:\ost>git remote -v

C:\ost>git remote add origin <https://github.com/chandanktech1/OST-Chandan>

C:\ost>git remote -v

origin <https://github.com/chandanktech1/OST-Chandan> (fetch)

origin <https://github.com/chandanktech1/OST-Chandan> (push)

C:\ost>git push -u origin main

error: src refspec main does not match any

error: failed to push some refs to '<https://github.com/chandanktech1/OST-Chandan>'

C:\ost>git push -u origin main

error: src refspec main does not match any

error: failed to push some refs to '<https://github.com/chandanktech1/OST-Chandan>'

C:\ost>git pull --rebase origin main

remote: Enumerating objects: 9, done.

remote: Counting objects: 100% (9/9), done.

remote: Compressing objects: 100% (7/7), done.

remote: Total 9 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)

Unpacking objects: 100% (9/9), 331.46 KiB | 341.00 KiB/s, done.

From <https://github.com/chandanktech1/OST-Chandan>

* branch main -> FETCH_HEAD

* [new branch] main -> origin/main

Successfully rebased and updated refs/heads/dev.

C:\ost>dir

Volume in drive C has no label.

Volume Serial Number is 95C3-F9A6

Directory of C:\ost

29-01-2026 12:49 <DIR> .

29-01-2026 12:49 <DIR> ..

29-01-2026 12:49 228,913 01CE0618 - Open Source Technologies Syllabus.pdf

29-01-2026 12:49 194,422 Practical-1.pdf

29-01-2026 12:49 13 README.md

29-01-2026 12:49 23 sample.txt

4 File(s) 423,371 bytes

2 Dir(s) 74,905,800,704 bytes free

C:\ost>git push origin main

error: src refspec main does not match any

error: failed to push some refs to 'https://github.com/chandanktech1/OST-Chandan'

C:\ost>git push origin main

error: src refspec main does not match any

error: failed to push some refs to 'https://github.com/chandanktech1/OST-Chandan'

C:\ost>git pull --rebase origin main

From <https://github.com/chandanktech1/OST-Chandan>

* branch main -> FETCH_HEAD

Current branch dev is up to date.


```
C:\ost>git push origin main

error: src refspec main does not match any

error: failed to push some refs to 'https://github.com/chandanktech1/OST-Chandan'

C:\ost>

C:\ost>git push origin main --force

error: src refspec main does not match any

error: failed to push some refs to 'https://github.com/chandanktech1/OST-Chandan'

C:\ost>git push -u origin main

error: src refspec main does not match any

error: failed to push some refs to 'https://github.com/chandanktech1/OST-Chandan'

C:\ost>git push -u origin dev

info: please complete authentication in your browser...

Enumerating objects: 7, done.

Counting objects: 100% (7/7), done.

Delta compression using up to 4 threads

Compressing objects: 100% (4/4), done.

Writing objects: 100% (6/6), 502 bytes | 251.00 KiB/s, done.

Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)

remote: Resolving deltas: 100% (2/2), completed with 1 local object.

remote:

remote: Create a pull request for 'dev' on GitHub by visiting:
remote:   https://github.com/chandanktech1/OST-Chandan/pull/new/dev
remote:

To https://github.com/chandanktech1/OST-Chandan
* [new branch]    dev -> dev

branch 'dev' set up to track 'origin/dev'.

C:\ost>

C:\ost>git push origin dev:main
```

Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

To <https://github.com/chandanktech1/OST-Chandan>

c7e9d82..847adc0 dev -> main

C:\ost>

C:\ost>local dev → remote main

'local' is not recognized as an internal or external command,
operable program or batch file.

C:\ost>

C:\ost>git checkout -b main

Switched to a new branch 'main'

C:\ost>git push -u origin main

branch 'main' set up to track 'origin/main'.

Everything up-to-date

C:\ost>git checkout main

Already on 'main'

Your branch is up to date with 'origin/main'.

C:\ost>git branch

dev

* main

master

C:\ost>* dev

'*' is not recognized as an internal or external command,
operable program or batch file.

C:\ost>git push -u origin dev

branch 'dev' set up to track 'origin/dev'.

Everything up-to-date

C:\ost>

C:\ost>

```
C:\ost>
C:\ost>git push origin dev:main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/chandanktech1/OST-Chandan
   c7e9d82..847adc0  dev -> main

C:\ost>
C:\ost>local dev -> remote main
'local' is not recognized as an internal or external command,
operable program or batch file.

C:\ost>
C:\ost>git checkout -b main
Switched to a new branch 'main'

C:\ost>git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date

C:\ost>git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

C:\ost>git branch
  dev
* main
  master

C:\ost>* dev
'*' is not recognized as an internal or external command,
operable program or batch file.

C:\ost>git push -u origin dev
branch 'dev' set up to track 'origin/dev'.
Everything up-to-date

C:\ost>
C:\ost>
```

github.com/chandanktech1/OST-Chandan

chandanktech1 / OST-Chandan

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

OST-Chandan Public

Pin Watch 0 Fork 0 Star 0

main 2 Branches 0 Tags

Go to file Add file Code

Students Update in dev 847adc0 · 20 minutes ago 5 Commits

01CE0618 - Open Source Technologies Syllabu...	Add files via upload	last week
Practical-1.pdf	Add files via upload	last week
README.md	Initial commit	last week
sample.txt	Update in dev	20 minutes ago

README

OST-Chandan

No description, website, or topics provided.

Readme Activity 0 stars 0 watching 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Experiment 1

AIM: Explore GitHub/GitLab for open-source projects with different licenses

1. List of GitHub Licenses

- MIT license
- Apache license 2.0
- GNU General Public license (GPL v2.0)
- GNU General Public license (GPL v3.0)
- GNU Lesser General Public license (LGPL)
- BSD 2-Clause license
- BSD 3-Clause license
- Mozilla Public license 2.0 (MPL 2.0)
- Eclipse Public license (EPL)
- Creative Commons licenses
- The Unlicense
- Proprietary / No license

2. Licenses and Short Description Table

No.	License	Short Description
1	MIT license	Very permissive; allows use, modification, and distribution with attribution.
2	Apache license 2.0	Permissive license with explicit patent protection.
3	GNU GPL v2.0	Strong copyleft; modified code must remain open source.
4	GNU GPL v3.0	GPL v2.0 plus protection against patent abuse and tivoization.
5	GNU LGPL	Allows linking with proprietary software; mainly for libraries.

No.	License	Short Description
6	BSD 2-Clause	Simple permissive license with minimal restrictions.
7	BSD 3-Clause	BSD 2-Clause plus restriction on using author's name for promotion.
8	Mozilla Public license 2.0	File-level copyleft; allows mixing open and closed source.
9	Eclipse Public license (EPL)	Weak copyleft; commonly used for enterprise software.
10	Creative Commons licenses	Used for documentation/media, not recommended for software.
11	The Unlicense	Places code in the public domain with no restrictions.
12	Proprietary / No license	Code cannot be legally used, modified, or shared by others.

3. Licenses Comparison Table

License	Commercial Use	Open Source Required	Patent Protection
MIT	Yes	No	No
Apache 2.0	Yes	No	Yes
GPL v2.0	Limited	Yes	No
GPL v3.0	Limited	Yes	Yes
LGPL	Yes	Partial	No
BSD 2-Clause	Yes	No	No
BSD 3-Clause	Yes	No	No
MPL 2.0	Yes	File-level	No
EPL	Yes	Partial	No

License	Commercial Use	Open Source Required	Patent Protection
Creative Commons	Limited	Depends	No
Unlicense	Yes	No	No
Proprietary	No	No	No

4. List of GitHub Alternatives

Platform	Short Description
GitLab	Complete DevOps platform with Git repositories and CI/CD.
Bitbucket	Git repository hosting integrated with Atlassian tools.
SourceForge	Hosting platform mainly for open-source projects.
Azure DevOps	Microsoft's service for Git repos, pipelines, and project management.
Gitea	Lightweight, self-hosted Git service.
Gogs	Simple and fast self-hosted Git repository solution.
AWS CodeCommit	Fully managed Git service by AWS.
Phabricator	Code review and collaboration tool for developers.
Beanstalk	Secure Git hosting with deployment features.
Launchpad	Development platform mainly used for Ubuntu projects.

5. GitHub vs GitLab Table

Feature	GitHub	GitLab
Owner	Microsoft	Gitlab Inc.
Type	Code hosting platform	Complete DevOps platform
CI/CD	GitHub Actions (external setup)	Built-in CI/CD
Self-Hosting	Limited	Fully supported
Open Source	Partially	Core features open source
Issue Tracking	Yes	Yes (more advanced)
DevOps Tools	Limited	End-to-end DevOps
Popularity	Most popular platform	Popular in enterprises
Free Private Repos	Yes	Yes
Best For	Open-source fi collaboration	Full DevOps lifecycle

6. Open-Source vs Proprietary vs Freeware

Feature	Open-Source Software	Proprietary Software	Freeware
Source Code	Available to users	Not available	Not available
Cost	Free (mostly)	Paid (usually)	Free
Modification	Allowed	Not allowed	Not allowed
Redistribution	Allowed	Restricted	Restricted
Ownership	Community / Organization	Company / Vendor	Company / Developer
Transparency	High	Low	Low
Examples	Linux, Firefox	Windows, MS Office	Adobe Reader, VLC
Best Used For	Learning, development, customization	Commercial / enterprise use	Personal use